

AVORD AB, France

WMO: 072570

Lat: 47.0522N

Lon: 2.6422E

Elev: 177

StdP: 99.22

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-3.8	-8.8	1.8	-3.9	-6.5	2.2	-2.3	12.7	9.5	11.4	9.0	3.2	40	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.8	20.4	30.2	19.8	28.1	19.0	21.4	29.6	20.6	28.2	19.7	26.7	3.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.1	23.9	18.1	13.3	22.7	17.2	12.6	22.1	62.8	29.9	59.8	28.3	57.1	26.8	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.7	7.7	-8.0	35.8	2.7	2.3	-9.9	37.5	-11.5	38.9	-13.0	40.2	-14.9	41.9		
			-8.4	23.2	2.6	0.9	-10.3	23.8	-11.8	24.4	-13.3	24.9	-15.2	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	4.1	4.7	7.9	10.7	14.4	18.4	20.1	19.9	16.5	12.9	7.5	4.7
	DBStd	6.85	4.09	3.96	3.62	3.35	3.44	3.47	3.27	3.47	3.15	3.84	3.93	4.11
	HDD10.0	736	186	152	83	30	3	0	0	0	1	18	94	169
	HDD18.3	2578	442	383	322	228	128	42	18	21	73	172	326	424
	CDD10.0	1412	3	2	19	52	142	251	313	306	194	107	18	4
	CDD18.3	212	0	0	0	1	7	42	73	69	17	3	0	0
	CDH23.3	2282	0	0	0	15	98	457	768	743	181	20	0	0
	CDH26.7	819	0	0	0	1	13	156	293	312	44	1	0	0
Wind	WSAvg	3.5	3.8	4.0	4.0	3.7	3.4	3.3	3.4	3.1	3.2	3.2	3.5	3.8
Precipitation	PrecAvg	825	65	59	55	69	76	61	72	68	64	79	80	77
	PrecMax	1055	155	112	153	208	181	117	180	163	164	141	167	160
	PrecMin	613	7	6	10	9	15	7	15	7	20	23	31	20
	PrecStd	114	32	30	35	42	34	27	39	36	36	30	34	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.1	21.2	25.2	28.2	34.1	35.1	37.3	30.8	26.0	18.8	14.8
		MCWB	10.9	10.6	12.6	16.0	19.1	21.3	20.8	20.8	18.5	17.6	14.3	11.5
	2%	DB	12.5	13.9	18.6	22.2	26.0	30.5	32.1	33.0	27.8	22.9	16.1	13.1
		MCWB	10.6	9.8	11.5	14.3	17.5	20.3	20.6	20.4	18.2	16.7	13.0	11.0
	5%	DB	11.2	11.9	16.2	19.9	24.0	28.1	29.9	30.1	25.2	20.3	14.7	11.9
		MCWB	9.6	8.9	10.7	13.4	16.9	19.4	19.8	19.6	17.2	15.8	12.5	10.3
	10%	DB	9.9	10.3	14.1	17.5	21.8	25.8	27.8	27.3	23.0	18.5	13.1	10.7
		MCWB	8.7	8.3	9.9	12.0	15.8	18.4	18.9	18.6	16.5	14.9	11.2	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	11.6	13.8	16.8	20.1	22.9	22.1	22.4	20.7	18.5	15.2	12.7
		MCDB	12.9	15.0	18.2	23.8	26.5	30.6	31.2	30.8	26.4	23.4	16.9	13.7
	2%	WB	11.1	10.5	12.5	15.1	18.7	21.3	21.1	21.1	19.3	17.3	13.7	11.5
		MCDB	12.3	12.6	16.7	21.3	24.8	28.4	29.9	29.6	25.1	21.2	15.7	12.8
	5%	WB	9.9	9.6	11.5	13.7	17.4	20.1	20.3	20.2	18.2	16.3	12.5	10.5
		MCDB	11.0	11.3	15.2	18.5	22.4	26.5	28.3	28.4	23.6	19.8	14.4	11.7
	10%	WB	8.7	8.6	10.5	12.5	16.1	19.1	19.4	19.3	17.2	15.3	11.4	9.3
		MCDB	9.7	10.3	13.6	16.6	20.8	24.6	26.4	26.3	21.9	18.1	13.1	10.5
Mean Daily Temperature Range	5% DB	MDBR	5.7	7.5	9.4	10.4	10.4	11.0	11.4	11.5	11.0	9.0	6.7	6.0
		MCDBR	6.4	10.1	13.2	14.6	14.0	15.0	15.2	16.2	14.7	11.3	8.4	7.0
		MCWBR	5.0	6.7	7.6	7.8	7.0	6.3	5.9	6.1	6.6	6.2	5.6	5.5
	5% WB	MCDBR	5.8	8.6	11.0	13.4	12.3	13.4	13.7	13.6	11.8	10.2	7.7	6.3
	MCWBR	5.1	6.2	6.8	7.5	6.6	6.1	5.5	5.6	6.0	6.0	5.5	5.2	
Clear-Sky Solar Irradiance	taub	0.315	0.335	0.363	0.396	0.399	0.410	0.404	0.397	0.365	0.358	0.329	0.312	
	taud	2.455	2.399	2.327	2.234	2.274	2.282	2.294	2.330	2.399	2.424	2.455	2.444	
	Ebn at Noon	773	829	855	857	866	856	855	847	844	790	747	728	
	Edh at Noon	67	87	108	129	129	129	126	117	99	84	67	61	
All-Sky Solar Radiation	RadAvg	1.05	1.87	3.13	4.40	5.17	5.94	5.74	5.02	3.87	2.39	1.27	0.89	
	RadStd	0.14	0.28	0.38	0.54	0.50	0.55	0.43	0.45	0.30	0.18	0.14	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page